

Private Answers, Public Losses

ChatGPT and the Decline of Stack Overflow

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Abstract

This paper studies whether ChatGPT is associated with reduced public participation on Stack Overflow. We use Stack Exchange Data Explorer data on monthly question activity from January 2018 through December 2024 across ten programming tags. The analysis focuses on account age at the time of posting, which captures how long users had been connected to Stack Overflow's public Q&A system. Average monthly questions fall after ChatGPT, and the decline is largest among accounts aged 8-30 days and 31-365 days. In the preferred tag $\times$ account-age specification, the 8-30 day group declines by about 50.6 percent more than accounts older than one year, while the 31-365 day group declines by about 45.7 percent more. The results suggest that private AI assistance may weaken entry into public knowledge production by reducing participation among users with weaker platform attachment.

1. Introduction

Stack Overflow activity fell after ChatGPT's release, and the timing of that fall is important to study because Stack Overflow activity is a form of public participation. The fall in question volume could reflect private AI substitution, platform maturity, changing moderation norms, shifts in developer habits, or the continuation of a decline that was already underway before November 2022. ChatGPT changed the outside option for programmers seeking help, since a user who once had to search Stack Overflow or ask a public question could now ask a private AI system and receive an immediate response. The economic question is whether that change in the outside option corresponds to a measurable change in public participation, and where in the user base any such change appears.

Stack Overflow creates value through public knowledge production. A question posted on the platform can be answered, corrected, voted on, edited, and found by future users facing a similar programming problem, while the same problem solved through ChatGPT may help the original user without producing public information for others to search and reuse. ChatGPT therefore changes the private cost of getting help while creating a possible wedge between private benefit and public value, since the user captures the speed and convenience of private AI assistance while the public archive may lose a question-answer pair that future users could have searched, corrected, or built on.

This paper asks whether ChatGPT is associated with reduced public participation on Stack Overflow and where the decline appears most strongly. The empirical focus is account age at the time of posting, which gives a measurable proxy for embeddedness in Stack Overflow because users with older accounts are more likely to have accumulated reputation, platform habits, and familiarity with the norms of public Q&A, while newer accounts are closer to the entry margin of the contribution system, where users are still deciding whether the platform is worth learning and returning to. Account age captures connection to Stack Overflow more directly than programming ability, since it measures how long the user had been linked to the platform at the time of posting. A decline concentrated among newer accounts would point to a different concern than a decline spread evenly across all groups, because it would suggest that private AI assistance weakened participation among users who had begun using Stack Overflow but had not yet developed much attachment

to it. The economic concern reaches beyond the immediate loss of questions, since Stack Overflow depends on a continuing flow of users who read, ask, learn norms, vote, edit, and eventually answer, and a thinner entry margin can therefore weaken the future supply of contributors as well as current question volume.

We use data from Stack Exchange Data Explorer covering Stack Overflow questions from January 2018 through December 2024 across ten programming tags, including Python, JavaScript, Java, C++, C#, TypeScript, PHP, R, Swift, and Go. We first examine changes in question activity by account-age group, then construct a tag $\times$ month $\times$ account-age panel that lets us test whether the account-age pattern survives after preserving variation across programming areas. The post-ChatGPT period begins in December 2022, the first full month after ChatGPT's release.¹ The main empirical result is that the decline is largest among accounts aged 8-30 days and 31-365 days at the time of posting, with the 8-30 day group declining by about 50.6 percent more than accounts older than one year and the 31-365 day group declining by about 45.7 percent more in the preferred tag $\times$ account-age specification. The 0-7 day group also declines relative to older accounts, although the estimate is smaller and statistically insignificant. The pattern fits the argument that ChatGPT affected the entry margin into public Q&A participation, with the largest relative declines appearing among users who were less embedded in Stack Overflow.

Our contribution is to move the analysis inside Stack Overflow. Earlier work has shown that Stack Overflow activity declined after ChatGPT using external comparisons, including restricted-access platforms and Reddit developer communities (del Rio-Chanona, Laurentsyeve, and Wachs, 2024; Burtch, Lee, and Chen, 2024). We add account-age heterogeneity within the platform itself, which lets us study who appears to reduce participation rather than only how much overall participation fell. Tag variation lets us preserve differences across programming areas without assuming in advance that one language is more substitutable by ChatGPT than another, and the account-age margin is useful because it lets us ask whether the decline was concentrated among users closer to entry into the public contribution system.

The paper supports a careful interpretation of the post-ChatGPT decline. The platform was already trending downward before ChatGPT, and the data do not observe individual ChatGPT use, so the evidence

¹ The post-ChatGPT period begins in December 2022, the first full month after ChatGPT's release.

cannot explain the full decline in Stack Overflow activity. The welfare meaning of the decline also depends on whether the missing questions would have produced reusable public knowledge or mostly duplicated existing material. Those limits narrow the claim while leaving the main economic concern in place. If private AI assistance reduces participation most among newer accounts, the effect may reach beyond the immediate loss of questions and weaken the pipeline that sustains Stack Overflow's public stock of programming knowledge.

2. Literature Review

This paper connects three economic arguments that come together in the Stack Overflow setting. The first treats digital technologies and artificial intelligence as changes in the cost of producing, finding, and using information. The second treats knowledge as a nonrival input whose private and social values can diverge. The third treats platforms and voluntary contribution systems as institutions that depend on network effects, reputation, and the continued entry of new users. Stack Overflow sits at the intersection of these arguments because it is a digital platform where unpaid users produce searchable programming knowledge for other users, and ChatGPT changes the outside option facing those users by making private programming help faster and easier to obtain.

The digital economics literature provides the basic cost argument. Goldfarb and Tucker (2019) argue that digital technologies reduce several economic costs, including search, replication, transportation, tracking, and verification costs, and that framework fits this setting because Stack Overflow and ChatGPT both reduce the cost of solving programming problems while doing so through different institutions. Stack Overflow lowers search and information costs by organizing public questions, answers, votes, and tags into a reusable archive, while ChatGPT lowers the private cost of getting an initial answer because the user can ask privately, receive a response immediately, and avoid the public effort of formatting a Stack Overflow question. Users still face verification costs, hallucination risk, and the need to decide whether the answer works, so the more precise claim is that ChatGPT lowers the private time and effort cost of getting a first-pass answer.

The economics of artificial intelligence literature makes the same point in a more specific setting. Agrawal, Gans, and Goldfarb (2019) describe recent AI progress as a quality-adjusted fall in the price of prediction, while Furman and Seamans (2019) argue that AI can raise productivity while changing

labor-market and organizational behavior. These papers are useful because they treat AI as a change in relative costs, and the relevant relative cost in our setting is the cost of public participation compared with private assistance. Before ChatGPT, a user with a programming problem often had to search existing posts or ask a public question. After ChatGPT, the same user could start with a private AI answer, which can improve private productivity while reducing the incentive to create a public record of the problem.

The knowledge-economy literature explains why this private choice may have social consequences. Jones (2019), writing about Romer's contribution to growth theory, argues that ideas are nonrival because one person's use of an idea does not prevent others from using it, while Jones and Tonetti (2020) make a similar point about data, showing how nonrival information can create increasing returns because the same resource can be used by multiple agents. Stack Overflow questions and answers have this nonrival structure once they are posted, since a programming answer that helps one user can also help future users who search for the same error, library issue, or implementation problem. The private user choosing between ChatGPT and Stack Overflow may consider speed, convenience, and answer quality, while the public value of adding another searchable question-answer pair is less likely to enter the private decision.

Khan's discussion of knowledge institutions helps make this argument more precise because Stack Overflow is a specialized contribution system with natural barriers to use. Khan (2020) argues that open knowledge systems may still require specialized skills, tacit knowledge, or learning by doing before users can benefit from them, and her discussion of contribution goods helps clarify that some knowledge is nonrival while remaining partly excludable in practice, since outsiders need enough background knowledge to use the commons. Stack Overflow fits that description, since a posted programming answer is searchable and reusable, while the reader still needs enough technical context to understand the code, adapt it, and judge whether it applies. The welfare interpretation therefore depends on whether a missing public question would have added usable knowledge for future users.

Stack Overflow depends on network effects because the value of asking, answering, reading, and voting rises with the number of active users, which connects directly to the platform-economics literature. Rochet and Tirole (2003) argue that two-sided platforms must bring multiple groups of users onto the

platform, while Katz and Shapiro (1985) explain how network externalities make the value of a product or system depend on the number of other users. Stack Overflow has this structure even though users do not pay prices in the standard way, since askers need answer-writers, answer-writers need questions and an audience, readers need searchable content, and voters help validate quality. When one part of the platform weakens, the value of participation can fall for the other groups as well, and a decline in newer question-askers can reduce opportunities for answer-writers, lower the visibility of reputation, and weaken the system through which users move from reading to asking, voting, editing, and answering.

The literature on open source contribution helps explain why unpaid users contribute at all, since Lerner and Tirole (2002) show that open source participation can be explained partly through career concerns, reputation, peer recognition, and delayed labor-market returns. Stack Overflow differs from open source software development, although the incentive logic is close enough to be useful, since users often contribute without direct payment because answers, votes, badges, and reputation can signal technical ability to other developers. That incentive depends on visibility, since a useful answer has greater value when many users can see it, vote on it, and treat it as a signal of expertise. If public participation declines, the audience for reputation also shrinks, which can weaken the nonpecuniary return to contributing.

The closest empirical work studies whether ChatGPT reduced Stack Overflow activity after its release. Del Rio-Chanona, Laurentsyeve, and Wachs (2024) compare Stack Overflow with restricted-access counterfactual platforms and estimate a decline in Stack Overflow activity after ChatGPT, framing the issue as a threat to digital public goods because private LLM interactions may reduce the production of public human-generated knowledge. Burtch, Lee, and Chen (2024) compare Stack Overflow with Reddit developer communities and find that Stack Overflow visits and question volumes declined after ChatGPT while Reddit developer communities did not show the same pattern, which suggests that ChatGPT affects developer communities differently depending on how much the platform depends on standalone technical Q&A compared with discussion, identity, or social interaction.

We build on this empirical work by moving the analysis inside Stack Overflow. The existing studies are strongest at showing a post-ChatGPT decline using external comparisons, while we examine where that

decline appears within the platform by using account age and programming tags. Account age gives a proxy for embeddedness in Stack Overflow because users with older accounts are more likely to have accumulated reputation, platform habits, and familiarity with Q&A norms, while tag-level variation lets us preserve differences across programming areas without assuming in advance that one language is more substitutable by ChatGPT than another. The account-age margin is useful because it lets us ask whether the decline was concentrated among users closer to entry into the public contribution system, which is the part of the contributor pipeline most relevant to the long-run health of a voluntary knowledge-production platform.

The literature points to a coherent economic argument. ChatGPT lowers the private cost of getting programming help, while Stack Overflow produces public knowledge through voluntary participation on a platform with network effects and reputation incentives. The potential externality comes from the gap between the private value of a fast AI answer and the public value of a searchable question-answer pair, and the empirical analysis tests this argument by asking whether the post-ChatGPT decline is concentrated among newer accounts. If newer users are especially likely to reduce participation, private AI assistance may be weakening the pipeline through which users enter and sustain a public programming knowledge system.

3. Empirical Analysis

This section presents the empirical evidence on Stack Overflow activity before and after ChatGPT. We first describe the data and then move from descriptive evidence on the size of the post-ChatGPT decline to crosstabulations that examine the account-age pattern across tag groups, and finally to regression evidence that tests whether the decline is concentrated among newer accounts after controlling for tag-level differences and common monthly shocks.

3.1 Data

The data come from Stack Exchange Data Explorer and cover Stack Overflow questions from January 2018 through December 2024.² We focus on ten programming tags, which are Python, JavaScript, Java, C++, C#, TypeScript, PHP, R, Swift, and Go. These tags give the sample enough volume to examine monthly changes while preserving variation across programming areas.

² Stack Exchange Data Explorer is a public query interface for Stack Exchange network data. Because the database is periodically refreshed, counts may differ slightly from live Stack Overflow totals.

The main unit of observation is a month by account-age tier; for each question, account age is measured as the difference between the question creation date and the account creation date. Accounts are grouped into five categories, which are 0-7 days, 8-30 days, 31-365 days, 1+ year, and Deleted or Unknown. Account age captures how long the user had been connected to Stack Overflow at the time of posting, which makes it a proxy for embeddedness in the platform. Older accounts are more likely to have accumulated reputation, platform habits, and familiarity with the norms of public Q&A, while newer accounts are closer to the entry margin of the contribution system.

We also construct a tag $\times$ month $\times$ account-age panel. This second dataset keeps the same account-age structure while adding tag-level variation, which lets the analysis test whether the account-age pattern survives after preserving differences across programming tags. A question with multiple included tags can appear under more than one tag in this version of the data, so the tag-structured panel is a tag-level activity dataset, while the aggregate account-age panel gives the closer count of questions across the included sample.

We focus on questions because tags are directly observed on question posts and because question posting is the clearest measure of public demand for Stack Overflow help. The answer side would add evidence on the supply of unpaid expert effort, but answers inherit tags from their parent questions³, which requires heavier joins that were difficult to recover reliably through Stack Exchange Data Explorer within the project timeline. The empirical analysis therefore focuses on the point where programming problems first enter the public record.

3.2 Descriptive Evidence

Table 1 reports the size of the post-ChatGPT decline in the sample. Average monthly questions fall from about 69,385 before ChatGPT to about 22,877 after ChatGPT, which is a large reduction in public question production. Stack Overflow activity was already declining before December 2022, so the pre-post comparison cannot carry the causal argument by itself, although the descriptive evidence is still useful because

³ In Stack Exchange data, answers are stored as posts but do not carry their own tags. Tags must be recovered by joining answer posts to their parent question posts, which makes a tag $\times$ account-age answer panel harder to construct reliably within the project timeline.

it shows the scale of the decline and the shift toward older accounts that the regressions later examine in more detail. Average account age rises from about 950 days before ChatGPT to about 1,470 days after ChatGPT, so the remaining questions increasingly come from older accounts, and that compositional shift fits the entry-margin argument because a public Q&A system depends on a continuing flow of newer users who search, ask, learn the norms, and become more active participants over time.

Question characteristics also change after ChatGPT. Average body length rises from about 1,885 characters to about 2,333 characters, while the average number of tags rises from 3.21 to 3.37. Average answer count falls from 1.25 to 1.02, and the accepted-answer rate falls from 47.1 percent to 38.6 percent. These patterns point toward longer and more specific remaining questions paired with weaker answer outcomes, while views, scores, and accepted answers require some caution because older posts have had more time to accumulate engagement.

Table 2 separates the decline by account age at the time of posting. The largest interpretable declines are concentrated among accounts aged 8-30 days and 31-365 days, with average monthly questions from 8-30 day accounts falling by 79.3 percent and average monthly questions from 31-365 day accounts falling by 77.3 percent. The 1+ year group also declines sharply, although by a smaller 61.5 percent, and the 0-7 day group falls by 65.3 percent. Deleted or Unknown accounts fall by 85.4 percent, although that category is harder to interpret because the user account information is missing.

Figure A1 presents the same pattern using indexed activity, where each account-age tier is normalized to its own pre-ChatGPT average. This comparison helps because the 1+ year group has many more questions in levels, while the indexed figure compares proportional changes across groups. The 8-30 day and 31-365 day series fall the most after December 2022, which fits the argument that recently joined users were especially likely to reduce public Q&A participation after private AI assistance became available.

Figure A2 shows the monthly share of questions by account-age tier. The composition of remaining activity moves toward older accounts after ChatGPT, which supports the participant-pipeline interpretation because the decline appears in both question levels and the structure of remaining participation. If newer users become less likely to post, Stack Overflow loses current question volume while also weakening part of

the pathway through which users learn public Q&A norms, accumulate reputation, and become more active contributors.

3.3 Crosstabulations

The crosstabulations provide a direct way to examine the account-age pattern before turning to regression controls. Table 3 compares account-age changes across two tag groups. The top-three pre-period tags are Python, JavaScript, and Java, while the other included tags are C++, C#, TypeScript, PHP, R, Swift, and Go. This table helps show whether the account-age pattern is mainly coming from the largest language communities or whether it appears more widely across the sample.

The same account-age pattern appears in both tag groups. Among the other included tags, average monthly questions fell by 79.8 percent for 8-30 day accounts and 76.6 percent for 31-365 day accounts, compared with 58.7 percent for accounts older than one year. Among the top-three pre-period tags, average monthly questions fall by 79.1 percent for 8-30 day accounts and 77.6 percent for 31-365 day accounts, compared with 63.3 percent for accounts older than one year. The decline among newer accounts is therefore visible across the tag structure of the sample rather than being driven by a single language community.

Table 4 examines whether the composition of questions also changes by account-age group. Average body length rises for every account-age tier after ChatGPT. Accepted-answer rates fall for every account-age tier, and average answer counts also fall across all four interpretable groups. The share of questions with code blocks is relatively stable, while the average number of tags rises in every group. These patterns are consistent with a shift toward longer and somewhat more detailed questions combined with weaker answer outcomes, although the table leaves several possible interpretations open, including harder remaining questions, fewer available answer-writers, and changes in the kinds of users still posting.

3.4 Regression Evidence

The regression analysis tests whether younger account-age groups declined more than accounts older than one year after ChatGPT. The preferred specification uses the tag $\times$ month $\times$ account-age panel, which keeps the analysis inside the tag structure of Stack Overflow and controls for persistent differences between each tag and account-age group. The model is written as

$$\log(1 + \text{Questions}_{igt}) = \alpha_{ig} + \gamma_t + \beta(\text{Post}_t \times \text{AccountAgeTier}_g) + \varepsilon_{igt}$$

In this equation, i indexes the programming tag, g indexes the account-age group, and t indexes month. The term α_{ig} is a tag $\times$ account-age fixed effect, which absorbs persistent differences between groups such as Python questions from 1+ year accounts or Java questions from 8-30 day accounts. The term γ_t is a month fixed effect, which absorbs shocks common to all groups in the same month. The omitted comparison group is accounts older than one year at the time of posting. The month fixed effects absorb the common post-ChatGPT decline, so the interaction coefficients measure whether younger account-age groups changed differently from the 1+ year group after ChatGPT, with negative interaction coefficients meaning that the younger group declined more than the older-account comparison group.

Table 5 shows that the post-ChatGPT decline is concentrated among newer accounts. In the tag $\times$ account-age model, accounts aged 8-30 days declined by about 50.6 percent more than accounts older than one year, while accounts aged 31-365 days declined by about 45.7 percent more than accounts older than one year. Both estimates are statistically significant at the 1 percent level. The 0-7 day estimate is also negative, although it is smaller and statistically insignificant.

The aggregate account-age model gives a similar pattern, with the 8-30 day group and the 31-365 day group declining more than the 1+ year group after ChatGPT. The tag-structured model is the more important specification because it preserves variation across programming tags and still produces large, statistically significant estimates for the two recently joined groups, which makes the evidence stronger because the main account-age pattern remains after the model accounts for persistent tag-by-account-age differences.

The regression results point to embeddedness as the main account-age argument. Account age captures how long the user had been connected to Stack Overflow at the time of posting, which matters because newer accounts usually have less accumulated reputation, weaker platform habits, and less experience with the norms of public Q&A. The largest relative declines appear in the 8-30 day and 31-365 day groups, which are the users who had entered the platform but had not been there long enough to become deeply attached to it. The pattern fits the argument that ChatGPT affected the entry margin into public participation,

with the decline showing up most strongly among users who had the least reason to keep posting publicly once private AI assistance became available.

4. Discussion

The account-age result points to the entry margin of Stack Overflow participation, since newer users are the group most likely to still be deciding whether the platform is worth learning and returning to. Accounts aged 8-30 days and 31-365 days had already entered the platform enough to post, although they were much less embedded than users with accounts older than one year, since they likely had less accumulated reputation, weaker platform habits, and less familiarity with the norms of asking public technical questions. The larger post-ChatGPT decline among these groups fits the argument that private AI assistance weakened the incentive to keep entering Stack Overflow's public contribution system, especially for users who had not yet built much attachment to the platform.

The pipeline argument follows directly. A user can begin as a reader, then ask questions, receive answers, learn the norms of the platform, vote, edit, and eventually answer other users' questions, and the public value of Stack Overflow comes from this repeated movement across roles, since each stage helps maintain the archive and the community that produces it. A decline among newer accounts therefore affects more than the number of questions posted in a given month, because it can also weaken the path through which users become contributors to the public stock of programming knowledge.

The welfare meaning of the decline depends on what kind of public knowledge is being lost. Some missing questions were likely duplicates, narrow debugging cases, or questions that the existing archive already answered well, and a private ChatGPT response to those questions saves the user time while costing the public archive little. The concern is stronger when the missing questions would have clarified a real technical issue, produced a reusable answer, or helped draw a newer user further into the contribution system. The evidence shows a patterned decline in public question production, while the social value of the missing questions depends on what kind of knowledge would have been created.

The externality argument should stay tied to the observed account-age pattern. The economic issue is the wedge between the private value of getting a fast answer from ChatGPT and the public value of

producing a searchable question-answer pair, since a private AI exchange can solve the original user's problem while a Stack Overflow post can also help future users who search, evaluate, reuse, and improve the answer. The account-age result gives this welfare concern a sharper form because the decline appears strongest among users closer to the beginning of the participation pipeline, which is the part of the system most directly tied to whether the public archive continues to grow.

Stack Overflow is also a specialized contribution system, which makes the public-good argument more precise. Once a useful question and answer are posted, they are searchable and reusable, although their value depends on the user's technical context, since a programming answer often requires knowledge of syntax, libraries, debugging details, and the user's own codebase. The social value of a post therefore depends on whether it adds knowledge that future users can actually use, especially in cases where the issue was not already well covered in the archive.

The platform-economics literature gives further structure to the account-age result, since Stack Overflow relies on participation from askers, answer-writers, readers, and voters. If fewer newer users ask questions, answer-writers face a smaller or less active audience, and if answer-writers expect fewer readers, votes, and reputation gains, their own incentive to contribute can weaken as well. Reputation has value because it is visible to an audience that can evaluate it, and that visibility shrinks when public participation falls. Even though we measure questions directly, the platform logic makes the answer side relevant for interpreting the question-side decline, since the question and answer sides reinforce each other through the shared audience.

The Reddit comparison helps define the substitution margin without turning the discussion into a study of every developer platform. Burtch, Lee, and Chen (2024) compare Stack Overflow with Reddit developer communities after ChatGPT and find declines in Stack Overflow visits and question volume without the same pattern in Reddit developer communities, which they interpret as evidence that communities with more social fabric may be less exposed to AI substitution. For this paper, the comparison suggests that ChatGPT may substitute more directly for standalone technical Q&A than for online spaces where users also seek discussion, identity, and social interaction.

GitHub serves as a second comparison because it overlaps with Stack Overflow as a developer platform while organizing activity differently. GitHub structures activity around repositories, issues, commits, and collaboration rather than around standalone Q&A artifacts, which makes Stack Overflow a setting where ChatGPT can substitute for the platform's main output more directly. ChatGPT can answer technical questions, while the platform activity it replaces most directly is likely to be standalone public Q&A rather than activity tied to a specific repository or production codebase. This distinction supports a careful interpretation of the empirical result, since Stack Overflow is exposed to AI substitution on a margin that is harder to find on platforms organized around collaboration and project context.

These comparisons connect to the main empirical result. ChatGPT is most likely to substitute for Stack Overflow when the user wants a fast technical answer that can be separated from social interaction, reputation building, or project-specific collaboration, and Stack Overflow is exposed on that margin because many questions are organized around reusable technical problems. Reddit and GitHub help show why the same AI tool may affect different developer platforms differently, depending on whether users are mainly seeking an answer, a discussion, a visible contribution, or collaboration around a project.

Several limits follow from the design. The data observe Stack Overflow questions rather than individual ChatGPT use, so the analysis cannot identify the exact outside option chosen by each user, while Stack Overflow was also declining before November 2022, which means timing alone cannot carry the causal argument. Account age captures embeddedness in Stack Overflow rather than the user's outside programming experience, since a newer account may still belong to an experienced programmer who had not used Stack Overflow much before. The sample also focuses on ten programming tags, so the estimates describe a structured set of major programming communities rather than the full Stack Overflow platform.

The focus on questions creates one more boundary around the interpretation. Questions are the point where a programming problem first enters the public record, which makes them a useful measure of public demand for Stack Overflow help, and the answer side is also central because Stack Overflow's value depends on unpaid expert effort. Answers need to be joined to parent questions before tags can be assigned, and that tag-by-account-age answer panel was difficult to recover reliably through Stack Exchange Data

Explorer within the project timeline. The empirical analysis therefore focuses on public question production while leaving the answer-side question for later work.

Within these limits, the main interpretation is that ChatGPT may have weakened entry into Stack Overflow's public contribution system. The evidence is strongest for accounts that were recently created but had existed long enough to move beyond the first few days of use, since those users were still deciding whether Stack Overflow was worth learning and returning to, and private AI assistance may have reduced the value of posting publicly most for users with weaker platform attachment. If that interpretation is right, the public cost is a thinner pipeline of future contributors to a specialized knowledge system that depends on voluntary participation.

5. Conclusion

This paper studies whether ChatGPT is associated with reduced public participation on Stack Overflow, with account age used to examine where the decline appears most strongly. Stack Overflow matters economically because it is a public contribution system, where questions, answers, votes, and edits create a searchable archive that future users can reuse. ChatGPT changes the private cost of getting programming help, since users can ask privately, receive an immediate response, and avoid the effort of formatting a public question. The economic concern is that this private substitution may reduce the production of public programming knowledge.

The evidence supports a cautious interpretation, since Stack Overflow activity was already falling before December 2022, and the data does not observe individual ChatGPT use. Thus, the results describe where the post-ChatGPT decline is concentrated, nothing remotely close to a complete causal accounting of the platform's decline. Average monthly question volume falls after ChatGPT, and the decline is largest among accounts aged 8-30 days and 31-365 days at the time of posting. In the preferred tag $\times$ account-age specification, the 8-30 day group declines by about 50.6 percent more than accounts older than one year, while the 31-365 day group declines by about 45.7 percent more.

The pattern fits the argument that ChatGPT weakened the entry margin into public Q&A participation. Account age gives a useful measure of embeddedness in Stack Overflow because older accounts

are more likely to have platform habits, accumulated reputation, and familiarity with public Q&A norms, while newer accounts have weaker attachment to the platform and therefore had the least reason to keep posting publicly once private AI assistance became available. The result speaks to the process through which users enter and remain inside a public contribution system rather than to differences in programming skill across user groups.

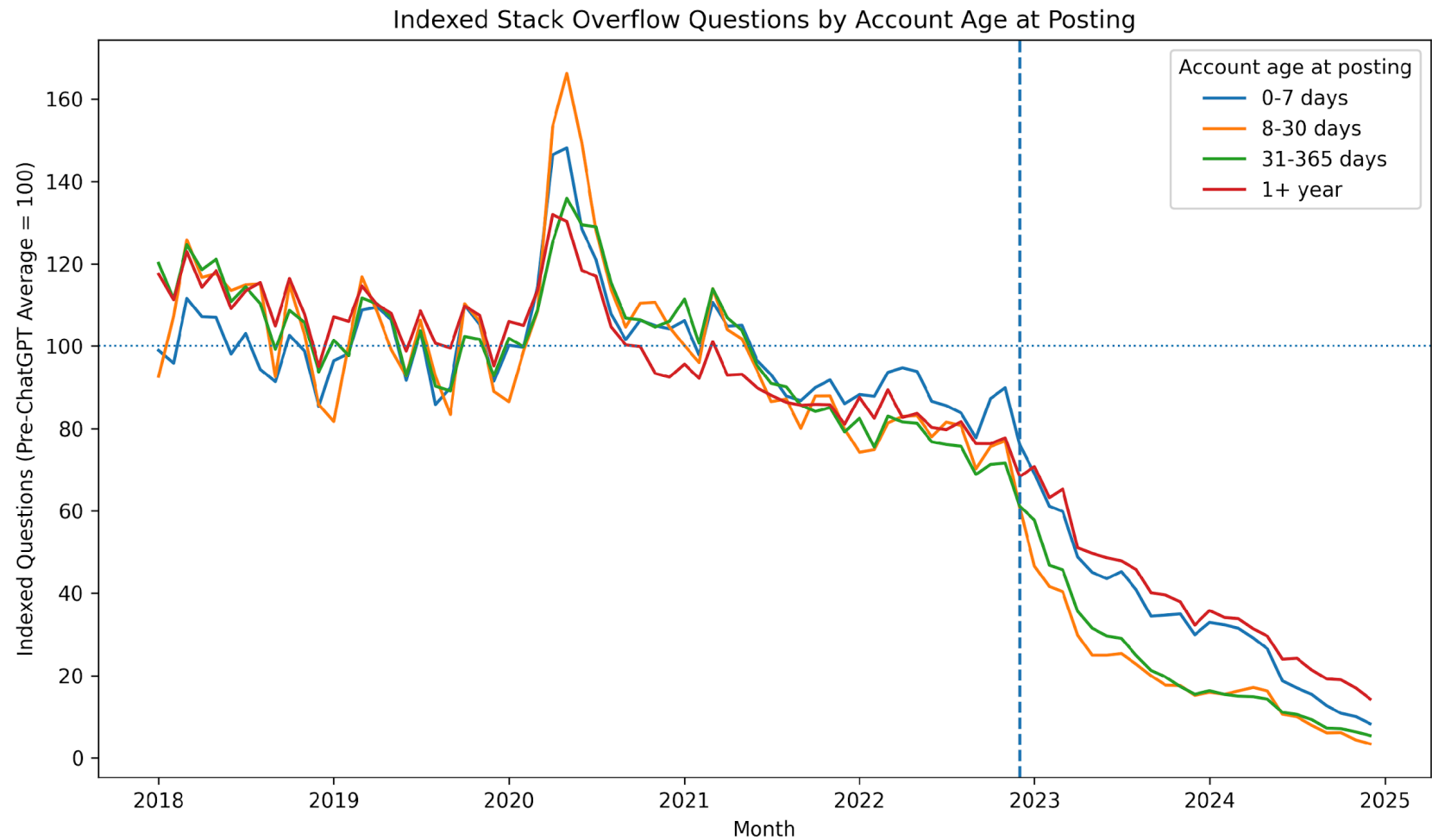
The welfare interpretation depends on the value of the missing questions. Some questions that shifted away from Stack Overflow were likely duplicates or narrow cases that ChatGPT could answer privately with little loss to the archive, while the concern is stronger when missing questions would have produced reusable answers, clarified technical problems, or helped newer users become future contributors. Stack Overflow's public value depends on repeated participation over time, so a decline among newer accounts can matter beyond the immediate loss in question volume.

Our contribution is to move the analysis inside Stack Overflow by focusing on who appears to reduce participation. Earlier work has documented a post-ChatGPT decline in Stack Overflow activity using external comparisons, while we add account-age heterogeneity and tag-structured evidence within the platform itself. The decline is concentrated among users closer to the entry margin and remains visible after preserving variation across programming tags, which is consistent with private AI assistance weakening public knowledge production by thinning the pipeline of future contributors to Stack Overflow.

The larger argument is that AI substitution may affect online knowledge systems unevenly, since a private AI answer can be valuable for the individual user, especially when the question is routine or already answered elsewhere, while the same private answer contributes less to a public archive that future users can search, evaluate, and reuse. Stack Overflow is useful for studying this tradeoff because its economic value comes from voluntary public contributions that accumulate over time. If newer users become less likely to enter that system, ChatGPT may affect both current question volume and the future supply of users who sustain the platform's stock of public programming knowledge.

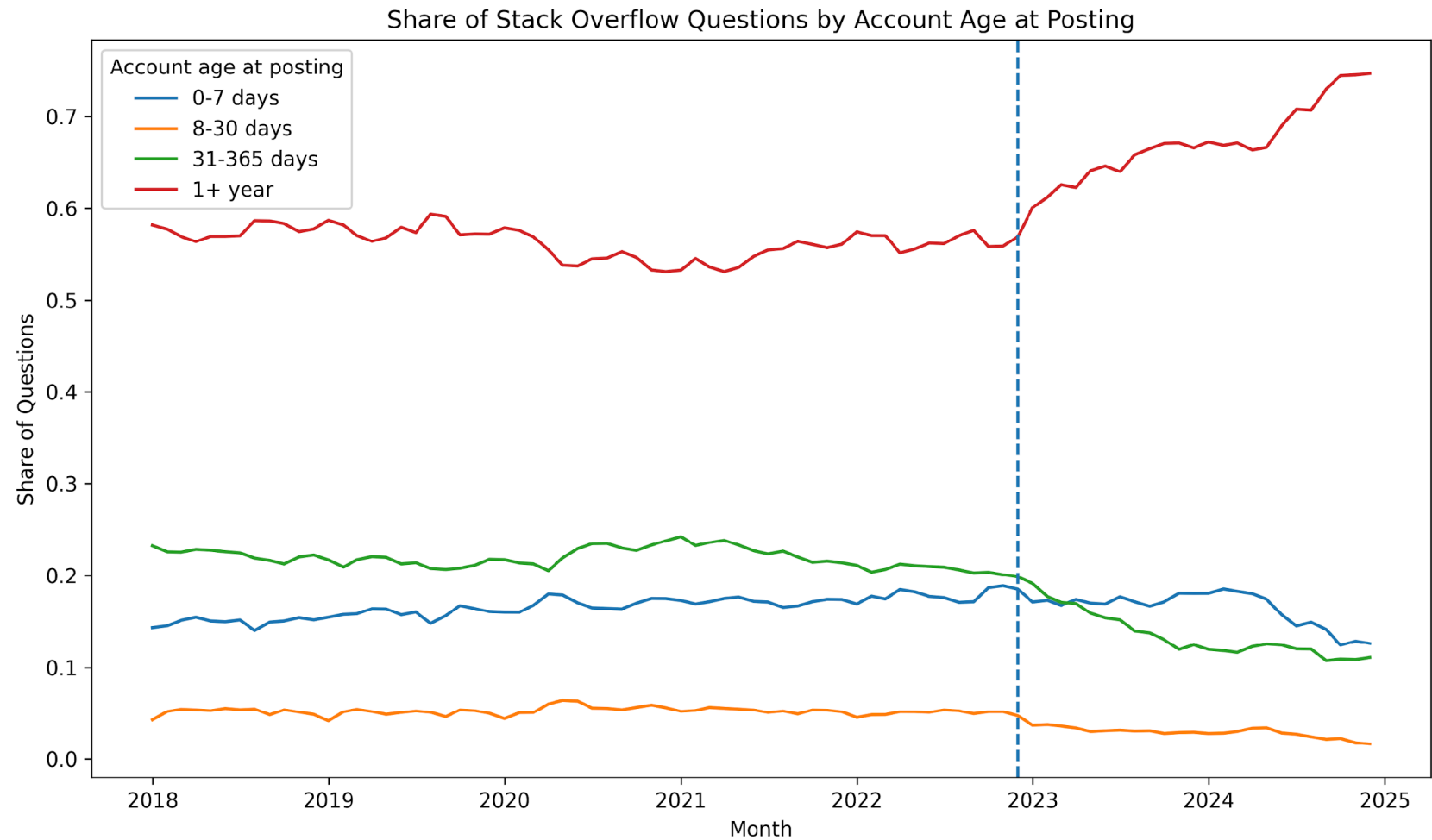
Appendix A. Figures

Figure A1. Indexed Stack Overflow Questions by Account Age at Posting.



Notes: Each account-age tier is indexed to its own pre-ChatGPT average, with January 2018 through November 2022 equal to 100. The vertical line marks December 2022, the first full month after ChatGPT's release. Account age is measured at the time the question was posted. Source: Stack Exchange Data Explorer.

Figure A2. Share of Stack Overflow Questions by Account Age at Posting.



Notes: The figure shows each account-age tier's monthly share of questions in the sample. Account age is measured at the time the question was posted. The vertical line marks December 2022, the first full month after ChatGPT's release. Source: Stack Exchange Data Explorer.

Figure A3. Sample of Tag x Account-Age Dataset.

TagName	MonthStart	AccountAgeTier	QuestionCount	AvgAccountAgeDays	AvgQuestionScore	AvgViewCount	AvgAnswerCount	AvgCommentCount	AcceptedAnswerRate	AvgTitleLength	AvgBodyLength	ShareWithCodeBlock	AvgNumTags	SourceFile	PostChatGPT	log_questions	Period	Tag group
c#	2018-01-01	0-7 days	1186.0	0.8760539629005060	0.4123102866779090	1163.066610455310	1.1770657672849900	3.1357504215851600	0.4080944350758850	55.33726812816190	1728.09190565469200	0.851602023608769	3.1863406408094400	tag_account_age_2018.csv	0	7.0791843948069670	Pre-ChatGPT	Other included tags
c#	2018-01-01	8-30 days	322.0	1.9844472049898900	0.2763975155279590	1161.9316770186300	0.062111801242240	0.4409937888198760	0.54590062111801200	54.590062111801200	0.8757763975155280	3.1335403276708100	tag_account_age_2018.csv	0	5.7776562323226600	Pre-ChatGPT	Other included tags	
c#	2018-01-01	31-365 days	1681.0	1.7616061867936600	0.6276026174895900	1488.180249851280	1.247471743010110	3.1760856632956600	0.47888616180084470	54.21475312314100	1814.1540749553800	0.8661511005353960	3.279005084840000	tag_account_age_2018.csv	0	7.427738340532890	Pre-ChatGPT	Other included tags
c#	2018-01-01	1+ year	5847.0	1.534.508294852060	1.4732348137164600	1974.9305426817200	1.2753548628459000	3.1364802462801400	0.5031640157345650	52.941944586967700	1822.7484179921300	0.8441936035573800	3.455618265777320	tag_account_age_2018.csv	0	8.673855001429620	Pre-ChatGPT	Other included tags
c#	2018-01-01	Deleted or Unknown	292.0		0.952054794520548	1530.0	1.3904109589041100	3.191780821917810	0.6198630136986300	52.01027397260270	1922.335616438360	0.910958904109589	3.164383561643840	tag_account_age_2018.csv	0	5.680172609017070	Pre-ChatGPT	Other included tags
c#	2018-02-01	0-7 days	1009.0	0.7849355797819620	0.4499504459861250	1213.7938553022800	1.1873141724479700	3.23191278493558	0.4043607532210110	56.001982160555	1892.6352824578800	0.8622398414271560	3.0099108027750200	tag_account_age_2018.csv	0	6.91770506993510	Pre-ChatGPT	Other included tags
c#	2018-02-01	8-30 days	414.0	18.22705314000660	0.3961352657004830	1346.1739130434800	1.292270531400970	0.4903381642512080	0.519468590338160	1783.2028985507200	0.9009661835748790	3.0990338164251200	tag_account_age_2018.csv	0	6.028278530230700	Pre-ChatGPT	Other included tags	
c#	2018-02-01	31-365 days	1521.0	1.76.8929651545040	0.5075608152531230	1261.2445759368800	1.2287968441814600	3.17619868507560	0.4812623274161740	55.15318669165020	1712.0841551610800	0.8533859303090070	3.2669296515450400	tag_account_age_2018.csv	0	7.327780538421630	Pre-ChatGPT	Other included tags
c#	2018-02-01	1+ year	5338.0	15.13.402023229670	1.6004121393780400	2166.3756428999600	1.2440989134570000	3.1534282502810000	0.5176095916073440	56.353128512551500	1845.3746271618600	0.8669913825402770	3.430123641813410	tag_account_age_2018.csv	0	8.582793648500190	Pre-ChatGPT	Other included tags
c#	2018-02-01	Deleted or Unknown	252.0		1.0277777777777800	1713.452380952380	1.2857142857142900	3.1904761904761900	0.5595238095238100	53.62698412698410	1804.4444444444400	0.8928571428571430	3.29862852396825400	tag_account_age_2018.csv	0	5.53389848872752	Pre-ChatGPT	Other included tags
c#	2018-03-01	0-7 days	1158.0	0.8359240069084630	0.3713298791019000	1188.940414507770	1.199481865284970	3.3048359240069100	0.3747841105354060	56.03626943005180	1782.3609671848000	0.8488773747841110	3.025906735751300	tag_account_age_2018.csv	0	7.05531284339750	Pre-ChatGPT	Other included tags
c#	2018-03-01	8-30 days	449.0	17.4543429840908	0.6726057906458800	2118.532293986640	1.2271714922049000	3.425389755011140	0.4743875278396440	56.220489977728300	1759.369710467710	0.8552338530066820	3.08240545211580	tag_account_age_2018.csv	0	6.1092475827643700	Pre-ChatGPT	Other included tags
c#	2018-03-01	31-365 days	1782.0	175.90011223344600	0.5993265993265990	1479.116727833900	1.255892558922680	3.4051627384960700	0.4826038159371490	55.47923681257010	1781.341189674520	0.8602693602693600	3.2542087542087500	tag_account_age_2018.csv	0	7.48605261786314	Pre-ChatGPT	Other included tags
c#	2018-03-01	1+ year	6068.0	1534.8412986156900	1.344924192485170	1831.3749176005300	1.20385629531971	3.433256427158870	0.4886288727752140	57.26532630191170	1832.9418259723100	0.832335530652600	tag_account_age_2018.csv	0	8.710949125835860	Pre-ChatGPT	Other included tags	
c#	2018-03-01	Deleted or Unknown	272.0		0.7647058823529410	1461.1691176470600	1.2904411764705900	3.709558823529410	0.5735294117647060	51.25735294117650	1854.4632352941200	0.9080882352941180	3.422794117647060	tag_account_age_2018.csv	0	5.60947179518496	Pre-ChatGPT	Other included tags
c#	2018-04-01	0-7 days	1205.0	0.6763485477178420	0.5983402489626560	1337.9767834854800	1.1867219917012400	3.304564315352700	0.3717842323651450	56.0	1960.1095435684600	0.8473029045643150	3.0356846473029000	tag_account_age_2018.csv	0	7.095064377287130	Pre-ChatGPT	Other included tags
c#	2018-04-01	8-30 days	391.0	18.51150895140670	0.4501278772378520	1003.93094629156	1.10230179028133	3.2736578600205600	0.4654731457800510	53.189225831202050	1882.8797953964200	0.907928388746803	3.132992327365730	tag_account_age_2018.csv	0	5.971261839790460	Pre-ChatGPT	Other included tags
c#	2018-04-01	31-365 days	1685.0	159.78456972393800	0.6320474777448070	1380.3222551928800	1.2112759643916900	3.5347181008902100	0.4468842729970330	54.122848664688400	1752.9513535115700	0.8623145400593470	3.2094955489614200	tag_account_age_2018.csv	0	7.430114138561800	Pre-ChatGPT	Other included tags
c#	2018-04-01	1+ year	5604.0	1509.55585296217	1.3740185581727300	1764.9768022840800	1.194682369735900	3.2865810135617400	0.4801927194860810	57.03301213419990	1835.582976445400	0.8611705924339760	3.3433261955745900	tag_account_age_2018.csv	0	8.6314143350626	Pre-ChatGPT	Other included tags
c#	2018-04-01	Deleted or Unknown	288.0		0.7361111111111110	1635.392361111110	1.3125	3.4548611111111110	0.4652777777777780	53.052083333333300	1680.7222222222200	0.8715277777777780	3.208333333333300	tag_account_age_2018.csv	0	5.666426688112430	Pre-ChatGPT	Other included tags
c#	2018-05-01	0-7 days	1262.0	0.8304278922345480	0.4120443740095090	1117.947702060220	1.1806656101426300	3.3985736925515100	0.3866877971473850	54.486529318542	1721.6790808216000	0.8557844690966720	3.000792300269400	tag_account_age_2018.csv	0	7.141245122350490	Pre-ChatGPT	Other included tags
c#	2018-05-01	8-30 days	483.0	17.401656314699800	0.5672877846579089	1436.470496894440	1.269151138716360	3.2298136645968270	0.5010351668737100	53.34575569358180	1636.7577393751600	0.863354076372670810	3.099378881987580	tag_account_age_2018.csv	0	6.182084906716630	Pre-ChatGPT	Other included tags
c#	2018-05-01	31-365 days	1823.0	159.4306088864510	0.6489303346132750	1331.857926494790	1.223253633318700	3.439385628085570	0.444322545255074	55.43964289632470	1793.5085024684800	0.8688974218321450	3.2325836533187100	tag_account_age_2018.csv	0	7.508787170634280	Pre-ChatGPT	Other included tags
c#	2018-05-01	1+ year	5993.0	1515.787585516440	1.5127648923744400	1901.358918738530	1.235274486901390	3.3405639913232100	0.490071750375438	56.80577340230270	1867.568162856670	0.8611713665943600	3.3695978641748700	tag_account_age_2018.csv	0	8.69851424787661	Pre-ChatGPT	Other included tags
c#	2018-05-01	Deleted or Unknown	311.0		0.797427652733119	1276.1093247588400	1.2829581903569100	3.707395498392280	0.5241157556270100	53.67524115755630	1685.450160771700	0.887459807073955	3.12540192926045	tag_account_age_2018.csv	0	5.7430031878094800	Pre-ChatGPT	Other included tags
c#	2018-06-01	0-7 days	1137.0	1.0105540897097600	0.4212840809146880	1205.5021987689000	1.1653474054529500	3.2647317502156800	0.3790677220756380	55.215479331574300	1724.324538258580	0.830255057167986	3.091468777484610	tag_account_age_2018.csv	0	7.037027614686280	Pre-ChatGPT	Other included tags
c#	2018-06-01	8-30 days	422.0	17.30094786729860	0.547393364928930	1287.710904793300	1.16113740758290	3.417061611374410	0.45260663507109	51.540284360189600	1636.3507109004470	0.8615165876777250	3.10900473936490	tag_account_age_2018.csv	0	6.047372179046280	Pre-ChatGPT	Other included tags
c#	2018-06-01	31-365 days	1588.0	161.7078086423200	0.7506297229219140	1530.9061712846300	1.1983627204030200	3.213476070528970	0.4590680100755670	54.23110831234260	1822.1926954114100	0.8652392947103280	3.268261964735520	tag_account_age_2018.csv	0	7.370860166536720	Pre-ChatGPT	Other included tags
c#	2018-06-01	1+ year	5323.0	1583.0984407289100	1.4816832613188100	1865.23407785271500	1.1963178658651100	3.208904752958860	0.4925793725342850	55.0751662596280300	1872.7974826225800	0.8609806500009390	3.3725342851775300	tag_account_age_2018.csv	0	8.579980179515000	Pre-ChatGPT	Other included tags
c#	2018-06-01	Deleted or Unknown	273.0		0.8571428571428570	1483.820512820510	1.3956043956044000	4.168498168498170	0.5714285714285710	53.717948717948700	1630.5054945054900	0.912087912087912	2.9853479853479900	tag_account_age_2018.csv	0	5.6131281063880700	Pre-ChatGPT	Other included tags
c#	2018-07-01	0-7 days	1167.0	0.9383033419023140	0.50471293916024	1247.3341902313600	1.1508140531276800	3.2776349614395900	0.3496143958869800	55.9700056898030	1776.6538131962300	0.8483290488431880	3.159383033419020	tag_account_age_2018.csv	0	7.0630481633881700	Pre-ChatGPT	Other included tags
c#	2018-07-01	8-30 days	425.0	17.896470588235300	0.5811764705882350	1505.5670588235300	1.2141176470588230	3.2776470588235300	0.4611764705882350	55.054117647058800	1791.68	0.8564705882352940	3.192941176470590	tag_account_age_2018.csv	0	6.054439346269370	Pre-ChatGPT	Other included tags
c#	2018-07-01	31-365 days	1792.0	168.39732142857100	0.7293526785714290	1619.4112723214300	1.334263928571430	4.4642857142857140	0.5977877857142900	53.997787857142900	1707.1462053571400	0.8738839285714290	3.2237723214285700	tag_account_age_2018.csv	0	7.4916454739605130	Pre-ChatGPT	Other included tags
c#	2018-07-01	1+ year	5744.0	1558.6201253481900	1.431232590529250	1996.495299442900	1.2414603593314800	3.190285515320330	0.4956476323119780	57.560759052924800	1894.6415389972100	0.8661211699164350	3.3760445682451300	tag_account_age_2018.csv	0	8.656085190282860	Pre-ChatGPT	Other included tags
c#	2018-07-01	Deleted or Unknown	239.0		0.5606694560669460	1401.8828451882800	1.192468619246860	3.606694560669460	0.4937238493723850	51.55230125523010	1633.7573221757300	0.8870292887029290	3.2384937238493700	tag_account_age_2018.csv	0	5.480638923419190	Pre-ChatGPT	Other included tags
c#	2018-08-01	0-7 days	1121.0	0.8073148974130240	0.7261373773416590	1371.4014272970600	1.2167707404103500	3.0276538904638700	0.371989295270790	57.32917038358610	1730.279214986620	0.8331846565566460	3.146297948260480	tag_account_age_2018.csv	0	7.022968086082640	Pre-ChatGPT	Other included tags
c#	2018-08-01	8-30 days	444.0	18.50675675675680	0.5630630630630630	1381.9752252252300	1.2297297297297300	3.220720720720720	0.4527027027027030	55.06981981981980	1961.90909090909	0.8716216216216220	3.09684684646850	tag_account_age_2018.csv	0	6.09807428216624	Pre-ChatGPT	Other included tags
c#	2018-08-01	31-365 days	1687.0	162.1754593953760	0.6259632483698870	1360.9579134558400	1.2400711321873100	3.1802015411973900	0.4754001185536460	56.827								

Table 2. Average Monthly Questions by Account Age Before and After ChatGPT.

AccountAgeTier	Pre-ChatGPT	Post-ChatGPT	Percent change
0-7 days	11,093.5	3,853.6	-65.263
8-30 days	3,535.9	730.680	-79.335
31-365 days	14,819.7	3,369.0	-77.267
1+ year	37,972.5	14,637.4	-61.453
Deleted or Unknown	1,963.1	286.720	-85.395

Notes: Account age is measured at the time the question was posted. Percent change compares average monthly question counts in the post-ChatGPT period with the pre-ChatGPT period. Source: Stack Exchange Data Explorer.

Table 3. Average Monthly Questions by Tag Group, Account Age, and Period.

Tag group	AccountAgeTier	Post-ChatGPT	Pre-ChatGPT	Percent change
Other included tags	0-7 days	1,318.6	3,812.2	-65.411
Other included tags	8-30 days	245.800	1,217.3	-79.807
Other included tags	31-365 days	1,212.7	5,186.6	-76.618
Other included tags	1+ year	6,515.7	15,772.1	-58.688
Top-three pre-period tags	0-7 days	2,634.8	7,568.3	-65.187
Top-three pre-period tags	8-30 days	505.680	2,416.5	-79.074
Top-three pre-period tags	31-365 days	2,254.6	10,052.9	-77.573
Top-three pre-period tags	1+ year	8,566.8	23,326.5	-63.274

Notes: The top-three pre-period tags are the three included tags with the largest question counts before December 2022. In this sample, they are Python, JavaScript, and Java. Other included tags are C++, C#, TypeScript, PHP, R, Swift, and Go. Percent change compares average monthly question counts before and after ChatGPT's release. Source: Stack Exchange Data Explorer.

Table 4. Question Characteristics by Account Age and Period.

Account age at posting	Period	Average body length	Share with code block	Accepted answer rate	Average answer count	Average number of tags
0-7 days	Pre-ChatGPT	1,859.7	0.885	0.378	1.218	3.039
0-7 days	Post-ChatGPT	2,327.3	0.876	0.282	0.979	3.314
8-30 days	Pre-ChatGPT	1,891.5	0.912	0.458	1.283	3.097
8-30 days	Post-ChatGPT	2,369.8	0.909	0.361	1.045	3.311
31-365 days	Pre-ChatGPT	1,887.6	0.905	0.478	1.277	3.199
31-365 days	Post-ChatGPT	2,367.6	0.907	0.367	1.027	3.387
1+ year	Pre-ChatGPT	1,891.4	0.909	0.492	1.245	3.282
1+ year	Post-ChatGPT	2,326.2	0.921	0.417	1.031	3.380

Notes: The table reports weighted averages by question count. Account age is measured at the time of posting. These characteristics describe question composition and answer outcomes, not user skill. Source: Stack Exchange Data Explorer.

Table 5. Regression Results.

Variable	Model 1. Account-Age Panel	Model 2. Tag x Account-Age Panel
Post x 0-7 days	-0.173*** (0.044)	-0.133 (0.139)
Post x 8-30 days	-0.751*** (0.050)	-0.705*** (0.135)
Post x 31-365 days	-0.655*** (0.047)	-0.610*** (0.137)
Account-age fixed effects	Yes	No
Tag x account-age fixed effects	No	Yes
Month fixed effects	Yes	Yes
Standard errors	Robust	Clustered by tag x account-age
Observations	336	3,360
R-squared	0.996	0.976

Notes: The dependent variable is $\log(1 + \text{monthly questions})$. The omitted comparison group is accounts older than one year at the time of posting. Model 1 uses the aggregate account-age panel. Model 2 uses the tag x account-age panel. Negative interaction coefficients mean the younger account-age group declined more than the 1+ year group after ChatGPT. Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6. Correlation Matrix.

Variable	Questions	Account age	Question score	Views	Answers	Comments	Accepted answer rate	Title length	Body length	Code block share	Number of tags
Questions	1.000	0.685	0.754	0.493	0.138	-0.291	0.248	-0.127	-0.338	-0.171	0.147
Account age	0.685	1.000	0.567	0.140	-0.102	-0.024	0.546	0.159	0.058	0.372	0.460
Question score	0.754	0.567	1.000	0.748	0.506	0.175	0.563	-0.431	-0.372	0.104	-0.002
Views	0.493	0.140	0.748	1.000	0.769	0.105	0.589	-0.097	-0.642	-0.015	-0.403
Answers	0.138	-0.102	0.506	0.769	1.000	0.158	0.819	-0.871	-0.770	0.152	-0.682
Comments	-0.291	-0.024	0.175	0.105	0.158	1.000	0.154	-0.085	0.286	0.317	0.068
Accepted answer rate	0.248	0.546	0.563	0.589	0.819	0.154	1.000	-0.714	-0.583	0.446	-0.333
Title length	-0.127	0.159	-0.431	-0.097	-0.871	-0.085	-0.714	1.000	0.759	-0.035	0.598
Body length	-0.338	0.058	-0.372	-0.642	-0.770	0.286	-0.583	0.759	1.000	0.268	0.683
Code block share	-0.171	0.372	0.104	-0.015	0.152	0.317	0.446	-0.035	0.268	1.000	0.093
Number of tags	0.147	0.460	-0.002	-0.403	-0.682	0.068	-0.333	0.598	0.683	0.093	1.000

Notes: The matrix reports pairwise correlations across numerical variables in the account-age panel. Several engagement variables, especially views and scores, are affected by the age of the post. Source: Stack Exchange Data Explorer.

Appendix C. Data Source and Construction

The data come from Stack Exchange Data Explorer and cover Stack Overflow questions in ten programming tags from January 2018 through December 2024. The tags are Python, JavaScript, Java, C++, C#, TypeScript, PHP, R, Swift, and Go. The main account-age dataset aggregates questions by month and account-age tier, while the tag-structured dataset aggregates questions by tag, month, and account-age tier.

Account age is measured as the difference between the question creation date and the account creation date. The account-age groups are 0-7 days, 8-30 days, 31-365 days, 1+ year, and Deleted or Unknown. The post-ChatGPT period begins in December 2022, the first full month after ChatGPT's release.

We focus on questions because question tags are directly observed and because question posting is the clearest measure of public demand for Stack Overflow help.

Answer-side analysis would be valuable, but answers do not carry tags directly and must be assigned to the tags of their parent questions. Building that panel by tag and account age was difficult to complete reliably through Stack Exchange Data Explorer within the project timeline, so answer-side behavior is left for future work.

Appendix D. Bibliography

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Data Source

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